ChE 381 Evaporation Lab

Final Lab Report

Unit Operations Lab 1

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Group #2

Ameen Abuzir

Joshua Laforest

Javier Silva

Cesearo Dilosa

Executive Summary

The objective of this experiment was to evaluate the performance of a single-effect evaporator by concentrating sugar under different vacuum pressures. A 10% wt sugar solution was circulated through the evaporator at three different vacuum levels. At each setting, condensate and flow rate were measured to calculate the capacity and economy. Temperature data results demonstrate that increasing the pressure increased the boiling point of the solution, increasing the driving force for heat transfer. As a result, high vacuum pressures produced greater evaporation capacities due to the temperature discrepancies between condensing steam and boiling water. From the analysis, it was concluded that vacuum pressure plays a significant role in evaporator performance. High levels of vacuum pressure improved the water removal process, increasing the system capacity. Despite this, the economy for the evaporator was mostly below unity, expected for a single-effect evaporator. These findings highlight the limitations of single-effect evaporators in terms of steam utilization. Although simple to operate, multiple-effect evaporators would be more efficient to achieve steam economies beyond one, optimizing industrial utilization.

Introduction

In this experiment we will be using a single-effect evaporator to evaporate a sugar solution to evaluate the effects parameters such as vacuum pressure. The objective of running this experiment is to compare how changes of pressure will affect the temperature of the running evaporator, the economy of how much liquid is consumed and the amount of product that leaves the evaporator at different pressures. In order to conduct this evaporator experiment, the evaporator is fed a 10%wt solution into an evaporator. When running the evaporator, a constant amount of deionized water is maintained in the evaporator to make sure that the procedure is kept running. Water vapor from the boiling liquid is condensed and and measured to see how much comes out through a controlled time interval to find the economy and measured to see how much vapor is being removed at certain time to find the capacity of the evaporator at different pressures.. A vacuum pump will be connected to a loop, controlling the pressure and will be adjusted at different pressures in order to test the theory of change in pressure influences the parameters of evaporation. This investigation is crucial in the chemical engineering as it provides theory and data on using pressure to increase the amount of heat in a system. This helps with lowering the amount of work in applying heat to conduct in evaporation, leading it to be used in large scale operations to increase efficiency and lower energy cost in heating the evaporator. For example, when operating a distillation on a compound with a high Boiling point the increase of pressure helps lower the boiling point lowering the energy requirments.

Theory

The purpose of the unit operation, evaporation, is to concentrate a solution of a nonvolatile solute (e.g. sugar) and a volatile solvent (e.g. water). Evaporating portions of the solvent produces a more viscous liquid solution. This practice is commonly used in the food processing industry to increase product shelf life, reduce volume/weight, and increase value [4]. The mechanisms behind this operation involve heat exchange and lowering the boiling point of the solution. Heat exchange is typically done with hot steam and the cooler solvent (e.g. water). As these two fluids exchange heat, steam condenses and vapor forms from the solution. The amount of vapor formed depends on the temperature difference between the steam and the solution. Also, on the boiling point of the solution, which can be manipulated through a vacuum system. As the vapor forms and steam condensates, they're typically recycled back into the process to maintain constant flow rates and solution concentration while in the separator. The solution, which is the main product in this type of operation, is collected or continues along to the next phase of the process. To gauge the efficiency of an operation like this, the amount of water evaporated per amount of steam used can be measured. This is called the 'steam economy'. It gives a look into the overall performance. Which for a single-effect evaporator is around 0.9 to 0.98]. Another output variable to look at is 'capacity' which measures the mass flow rate of the vapor product that's collected. In our experiment, we collected a certain volume of steam condensate and timed how long it took. We did the same for the water product. These numbers plus the temperature difference between the two helped us calculate the heat transferred in the heat exchanger as well as the efficiency and performance of the operation.

Results

Data Analysis:

Solution composition (target 10 wt% sugar):

- Water = 2710.3 g
- Sugar = 290.7 g
- Total = 3001 g = 9.7 wt% sugar
- Bucket tare = 471.4 g

Vacuum = 23.0 inHg

Evaporation (water product):

- Run 1: 600 mL in 2.33 min
- Run 2: 300 mL in 2.15 min
- Run 3: 300 mL in 2.15 min

Steam-side condensate (heat-exchanger):

- Run 1: 400 mL in 2:33.05 = 2.5508 min
- Run 2: 200 mL in 1:19.49 = 1.3248 min
- Run 3: 100 mL in 0:37.40 = 0.6233 min

Temperatures:

1. 63 °C, 2) 62.0 °C, 3) 62.0 °C, 4) 97.0 °C, 5) 33.0 °C, 6) N/A, 7) 34.0 °C, 8) 112.0 °C

Vacuum = 16.0 inHg

Evaporation (water product):

- Run 1: 200 mL in 2.11 min
- Run 2: 210 mL in 2.07 min
- Run 3: 115 mL in 2.05 min

Steam-side condensate (heat-exchanger):

- Run 1: 100 mL in 0:41.00 → 0.6833 min
- Run 2: 200 mL in 1:17.98 → 1.2997 min
- Run 3: 100 mL in 0:43.10 → 0.7183 min

Temperatures:

1. 73.0 °C, 2) 72.0 °C, 3) 30.0 °C, 4) 100.0 °C, 5) 38.0 °C, 6) N/A, 7) 42.0 °C, 8) 113.0 °C

<u>Vacuum = 10.0 inHg</u>

Evaporation (water product):

- Run 1: 95 mL in 2.07 min
- Run 2: 75 mL in 2.03 min
- Run 3: 100 mL in 2.08 min

Steam-side condensate (heat-exchanger):

- Run 1: 200 mL in 1:28.01 = 1.4668 min
- Run 2: 100 mL in 0:43.32 = 0.7220 min
- Run 3: 100 mL in 0:43.98 = 0.7330 min

Temperatures:

1. 87.0 °C, 2) 87.0 °C, 3) 24.0 °C, 4) 104.0 °C, 5) 45.0 °C, 6) N/A, 7) 53.0 °C, 8) 113.0 °C

Calculations & Results:

A) Evaporation (water product) rates

Vacuum (inHg)	Run	Volume (mL)	Time (min)	Rate (mL min^{-1})	Mass rate (kg h^{-1})

23.0	1	600	2.33	257.51	15.451
23.0	2	300	2.15	139.53	8.372
23.0	3	300	2.15	139.53	8.372
Avg: 23				178.86	10.732
16.0	1	200	2.11	94.79	5.687
16.0	2	210	2.07	101.45	6.087
16.0	3	115	2.05	56.10	3.366
Avg: 16				84.11	5.047
10.0	1	95	2.07	45.89	2.754
10.0	2	75	2.03	36.95	2.217
10.0	3	100	2.08	48.08	2.885
Avg:10				43.64	2.618

(Std. dev. of the mL min^{-1} rates: 23 inHg = 55.6; 16 inHg = 20.0; 10 inHg = 4.82.)

B) Steam-side condensate (steam used)

Vacuum (inHg)	Entry	Volume (mL)	Time (min)	Rate (mL min^{-1})	Mass rate (kg h^{-1})
23.0	1	400	2.5508	156.81	9.409
23.0	2	200	1.3248	150.96	9.058
23.0	3	100	0.6233	160.43	9.626

Avg: 23				156.07	9.364
16.0	1	100	0.6833	146.34	8.780
16.0	2	200	1.2997	153.89	9.233
16.0	3	100	0.7183	139.21	8.353
Avg: 16				146.48	8.789
10.0	1	200	1.4668	136.35	8.181
10.0	2	100	0.7220	138.50	8.310
10.0	3	100	0.7330	136.43	8.186
Avg: 10				137.09	8.226

C) Pressure conversion, temperatures, capacity & economy

Vacuum (inHg)	Absolute P (kPa)	T_{soln} (°C)	T_{steam} (°C)	ΔT (°C)	Capacity (kg h^{-1})	Steam used (kg h^{-1})	Economy (–)
23.0	23.44	62.5	97.0	34.5	10.732	9.364	1.15
16.0	47.14	72.5	100.0	27.5	5.047	8.789	0.574
10.0	67.46	87.0	104.0	17.0	2.618	8.226	0.318

Sample Calculations:

(Examples shown for the 23.0 inHg condition. Density assumed ρ = 1.00 kg/L.)

Convert vacuum (inHg) to absolute pressure (kPa):

Atmospheric pressure = 101.325 kPa. 1 inHg = 3.38639 kPa.

P_{abs} = 101.325 − (23.0 inHg)(3.38639 kPa/inHg) = 101.325 − 77.887 = 23.44 kPa.

Temperature driving force:

$\Delta T = T\_steam - T\_solution = 97.0\ °C - 62.5\ °C = 34.5\ °C$.

Evaporation rate for one run (example: 600 mL collected in 2.33 min)

Volumetric rate: $rV = 600\ mL / 2.33\ min = 257.51\ mL \cdot min^{-1}$.

Mass rate ($\rho = 1.00\ kg/L$):

$\dot{m} = rV \times (1\ L / 1000\ mL) \times (60\ min / h) \times (1.00\ kg / L)$

$= 257.51 \times 60 / 1000 = 15.451\ kg \cdot h^{-1}$.

Averaged capacity at 23.0 inHg (three runs):

Average volumetric rate: $rV = (257.51 + 139.53 + 139.53) / 3 = 178.86\ mL \cdot min^{-1}$.

Average mass rate (capacity): $m = 178.86 \times 60 / 1000 = 10.732\ kg \cdot h^{-1}$.

Steam used for one condensate entry:

Example: 400 mL in 2.5508 min → $rV = 400 / 2.5508 = 156.81\ mL \cdot min^{-1}$.

$\dot{m} = 156.81 \times 60 / 1000 = 9.409\ kg \cdot h^{-1}$.

Steam economy:

$E = (Capacity) / (Steam\ used) = 10.732 / 9.364 = 1.15$ (dimensionless).

Unit conversions used:

• 1 inHg = 3.38639 kPa → $P\_abs = 101.325 - (inHg \times 3.38639)$.

• 1 L = 1000 mL; 1 h = 60 min.

• Rate conversion: $(kg/h) = (mL/min) \times (60/1000) \times \rho\_(kg/L)$.

• A temperature difference in °C has the same numeric value in K (for ΔT only).

Error Analysis:

Quantitative uncertainty:

Assumptions for instrument precision
• Volume: ±2 mL for 200–400 mL cylinders; ±1 mL for 100 mL; ±5 mL for 600 mL beaker fills.
• Time: ±1 s (±0.017 min).
• Temperatures: ±0.5 °C (thermocouple).
• Vacuum gauge: ±2 inHg.
• Density: condensate/water taken as $\rho = 1.00\ kg/L$. This introduces a 2–4% systematic error because $\rho = 0.97–0.98\ kg/L$ near 90–100 °C.

Single-run rate uncertainty (example at 23.0 inHg, 600 mL in 2.33 min)
$r = V / t$. Assuming independent errors,
$(\sigma_r / r)^2 = (\sigma_V / V)^2 + (\sigma_t / t)^2$.
Using $\sigma_V = 5$ mL, $\sigma_t = 0.017$ min, $V = 600$ mL, $t = 2.33$ min:
$\sigma_V / V = 5/600 = 0.0083$; $\sigma_t / t = 0.017/2.33 = 0.0073$.
Combined relative uncertainty = $\sqrt{(0.0083^2 + 0.0073^2)} = 0.011$ (1.1%).
With $r = 257.5$ mL·min^{-1}, the absolute uncertainty is $\sigma_r = 2.8$ mL·min^{-1}, $r = 257.5 \pm 2.8$ mL·min^{-1} (= ±0.17 kg·h^{-1} after unit conversion).

Run-to-run scatter (random variability)
Standard deviations of the evaporation rates (mL·min^{-1}) across the three runs at each vacuum:
- 23 inHg: 55.6 mL·min^{-1}
- 16 inHg: 20.0 mL·min^{-1}
- 10 inHg: 4.82 mL·min^{-1}

This shows decreasing variability at higher absolute pressures. Steam-side rates varied much less (on the order of 5–7 mL·min^{-1}).

Uncertainty of the mean capacity and steam economy
Mean capacity and steam-use uncertainties were estimated using the standard error of the mean ($n = 3$) and error propagation for a ratio $E = C / S$ (assuming independence):
$(\sigma_E / E)^2 = (\sigma_C / C)^2 + (\sigma_S / S)^2$.
Numerical results used in the report:
- 23 inHg: $E = 1.15 \pm 0.25$ (22% relative).
- 16 inHg: $E = 0.574 \pm 0.098$ (17% relative).
- 10 inHg: $E = 0.318 \pm 0.025$ (7.8% relative).

These uncertainties are dominated by run-to-run scatter; instrument precision is a smaller contributor.

Qualitative error sources
- Steady-state assumption: Trials may include startup transients; true steady-state may not have been reached every time.
- Heat losses: Unmeasured external losses reduce apparent capacity and economy, especially at smaller ΔT.
- Vacuum/pressure calibration: ±2 inHg affects absolute pressure and boiling temperature, shifting ΔT.
- Density assumption: Using $\rho = 1.00$ kg/L for hot condensate/water slightly overestimates mass rates by 2–4%.
- Temperature non-uniformity: Solution and steam temperatures can vary spatially; single-point measurements under-represent gradients.
- Human read error: Meniscus reading and stopwatch reaction time.

Net effect

The trend is robust (capacity and economy increase with ΔT), but absolute values likely carry 10–25% combined uncertainty, largest at 23 inHg due to higher run-to-run variance.

Discussion

Evaporator behavior matched heat-transfer theory: deeper vacuum (lower absolute pressure) reduced the solution's boiling point and increased the steam–liquor driving force, so capacity rose while steam economy improved only modestly (single-effect limit). Averages across steady periods show capacity 10.73 - 5.05 - 2.62 kg h^{-1} and economy ≈1.15 → 0.57 → 0.32 as vacuum moved from 23 - 16 - 10 inHg (absolute pressure increasing). Measured ΔT tracked this trend (34.5 - 27.5 - 17.0 °C) and boiling temperature rose accordingly (62.5 - 72.5 - 87.0 °C), consistent with

$q = UA\Delta T$ and expected VLE/boiling-point elevation.

The single economy value >1 at deepest vacuum is attributable to method limits rather than physics: brief timing windows, imperfect synchronization between product and steam-condensate measurements, transient pressure/temperature drift before true steady state, density = 1.00 kg L^{-1} assumed for hot water. and gauge uncertainty. These biases readily shift economy by a few to several percent.

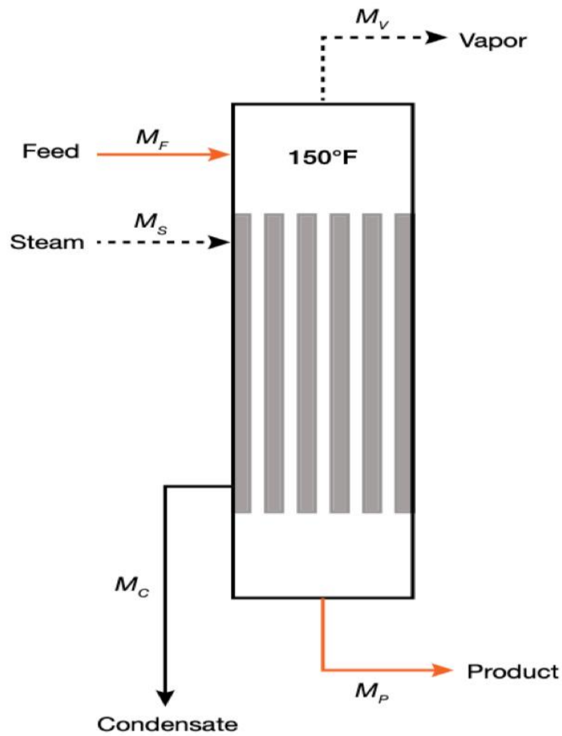
Recommended improvements: hold longer at steady state, time larger collection volumes, synchronize product and condensate intervals, log continuous P/T, and correct mass from temperature-dependent density. Bottom line: vacuum level is a strong lever for throughput in a single-effect unit; to raise steam economy meaningfully, a multiple-effect arrangement is required.

Team Contributions

	TIME (HOURS)	Team Member(s)
Operator (Both Lab Days)	3 Hours and 22 Mins	Ameen, Cesaro, Javier, Joshua
Pre-Lab Editing	1 hour	Ameen, Cesaro, Javier, Joshua
Experimental Objective (Prelab)	30 minutes	Joshua
Apparatus (prelab)	30 Minutes	Ameen

Materials & Supplies (prelab)	20 minutes	Joshua
Procedure (prelab)	50 min	Javier
Project Safety Evaluation (prelab)	1 hour	Ameen
Final Lab Editing	1 hour	Ameen, Cesaro, Javier, Joshua
Executive Summary (final lab)	40 minutes	Joshua
Introduction (Final lab)	40 min	Javier
Theory (Final Lab)	50 min	Cesaro
Results (final lab)	2 hours	Ameen
Discussion (final lab)	1 hour	Ameen
References (final and/or prelab)	50 min	Cesaro
Data Tabulation/Graphs (final)	1hour	Ameen
Error Analysis (final lab)	30 Minutes	Ameen
Sample Calculations (final Lab)	30 Minutes	Ameen
Power Point Presentation	1 hour	Joshua, Javier
Total	17 Hours and 32 Mins	

Energy Balance



▲ **Figure 3.** An energy balance for this evaporator assumes no work is performed on the system and heat loss is zero. To get a closer approximation of the evaporator rating, however, the heat loss is estimated to be 2% of the vapor energy input to each effect.

Evaporators are generally rated based on their evaporation rate — the amount of water they evaporate per hour (*e.g.,* kg/hr, ton/hr). The mass balances for the single-effect evaporator in Figure 3, which include the mass flowrates for feed (M_F), product (M_P), vapor (M_V), steam (M_S), and condensate (M_C), are:

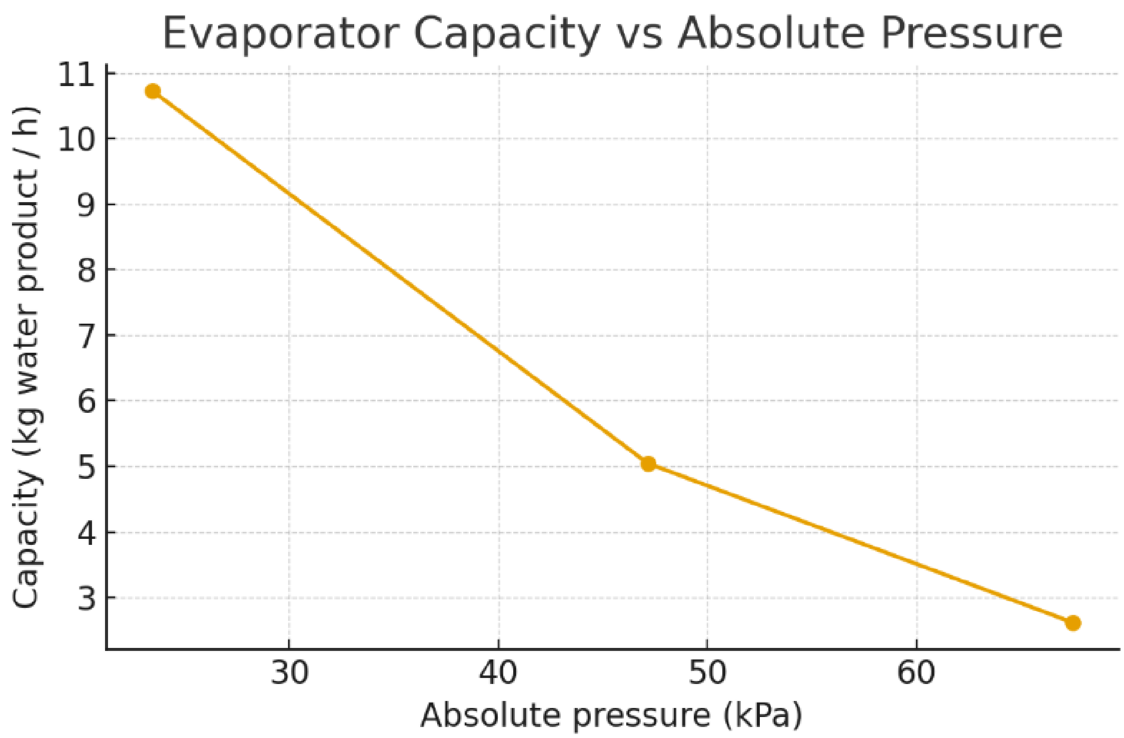
$$M_F = M_P + M_V \tag{1}$$
$$M_S = M_C \tag{2}$$

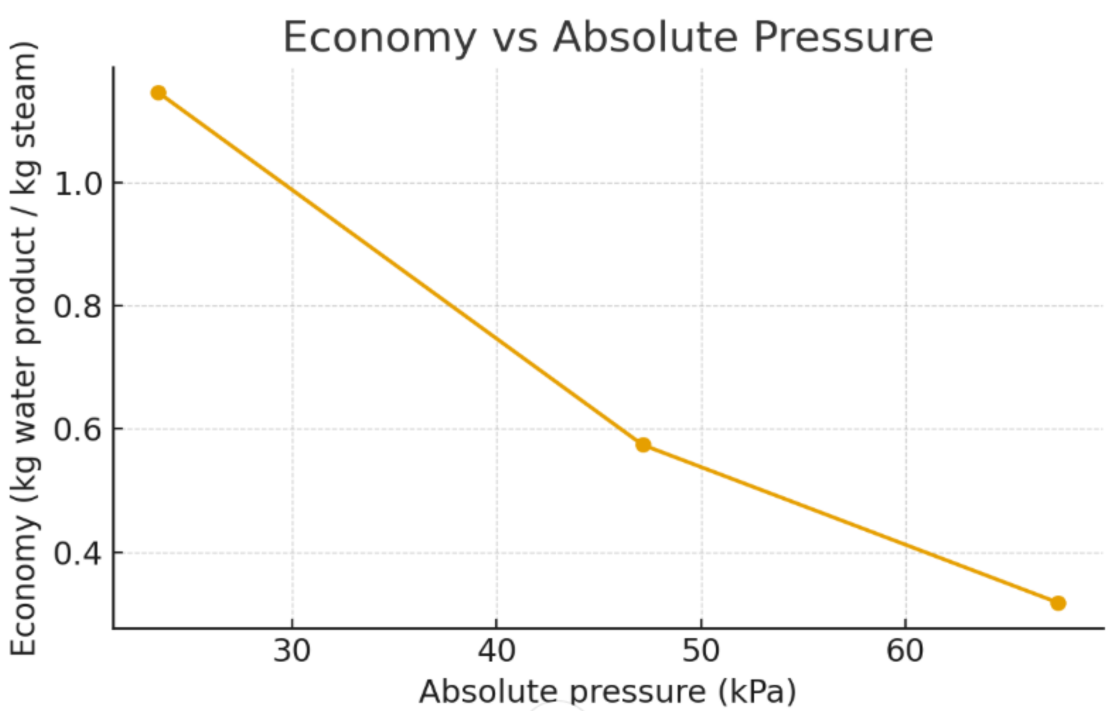
Drawing a control volume around the evaporator allows us to perform an energy balance of the system, in which h is the enthalpy of the respective streams:

$$M_F h_F + M_S h_s = M_P h_P + M_V h_V + M_C h_C \tag{3}$$

The energy balance assumes that no work is performed on the system ($W = 0$) and heat loss is negligible ($Q = 0$). In

Graphs





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